

Syllabus: Basics of Linear Algebra and Numerical methods

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1 General information

1.1 Time and place

There will be a total of 6 classes and a final exam. The first class of this course starts on September 12th, Saturday and the final exam will be given on October 24th, Saturday. The classes will start at 9:15 am and end at 11:45am. Each class will be divided in two parts, and each part will have its respective practice session. The venue will be classroom 2-502 in the Pázmány P. stny. 1C "Southern Building".

1.2 Materials for the class

The main material for the course will be the presentation slides presented each week and the jupyter notebooks. For books on the theoretical part I can recommend e.g., : [1].

Additionally, have installed Python together with the following libraries: Numpy, pandas, matplotlib, scipy, pillow (Python Image library), Jupyter notebooks and your favorite text editor or IDE.

1.3 Evaluation

The evaluation consist of two parts: homework assignments and a final written test.

Assignments: Each week will you will have to submit 3 exercises from a list of tasks. The deadline for this will always be 9 days after the lecture, i.e., until Monday. The maximum total score for each exercise is 3. Hence, the total score for this part is $6 \times 3 \times 3 = 54$.

Final exam: The final exam will be in person (mandatory). You are allowed to use the study material but no external help.

2 List of Topics

We summarize the topics presented each week:

- **Week 1:** Basic concepts in linear algebra: matrices, vectors, operations between them. Linear mappings and their relation with matrices. Brief overview of change of basis.
- **Week 2:** Generalizing vectors and matrices as multidimensional arrays. Array manipulation in Python with Numpy and the different methods found in this library. Brief overview of the chain rule in $\mathbb{R}$ and its generalization in multiple dimensions. Subspaces in $\mathbb{R}^d$ and linear independence: examples, subspaces generated by a set of vectors. Geometry in $\mathbb{R}^d$: orthogonal decomposition, length of a vector and dot product. More examples of special matrices.
- **Week 3:** Sparse matrices and its applications. Different structures of sparsity in Numpy. Linear systems, different techniques for solving them: Direct and iterative methods. The eigenvalue problem: geometrical interpretation and characterization. The power method: an iterative method for finding dominant eigenvalues.
- **Week 4:** A different way of looking at change of basis. Matrix decomposition for square and non-square matrices. The Singular Value decomposition: full and reduced form. Applications in compression. A brief overview of metric spaces, useful metrics in real-life applications.
- **Week 5:** A brief overview of Probability: important concepts and motivation. Normal distribution, transformation of random variables, standardization of random variables. Bivariate normal distribution. How to minimize multivariable functions, conditions for existence and uniqueness of minima. Finding local minimum, geometrical interpretation.
- **Week 6:** The gradient (steepest descent) method: description of the algorithm and main idea. Conditions for convergence and convergence rate. Methods for choosing parameters correctly (line search). Extensions of the gradient method; the conjugate gradient algorithm, Fletcher-Reeves method and their computational costs. The basic principles of the stochastic gradient method.

References

- [1] Gilbert Strang. *Linear algebra and its applications*. 2012.